Simulating Age-Period-Cohort Data

Volker Schmid

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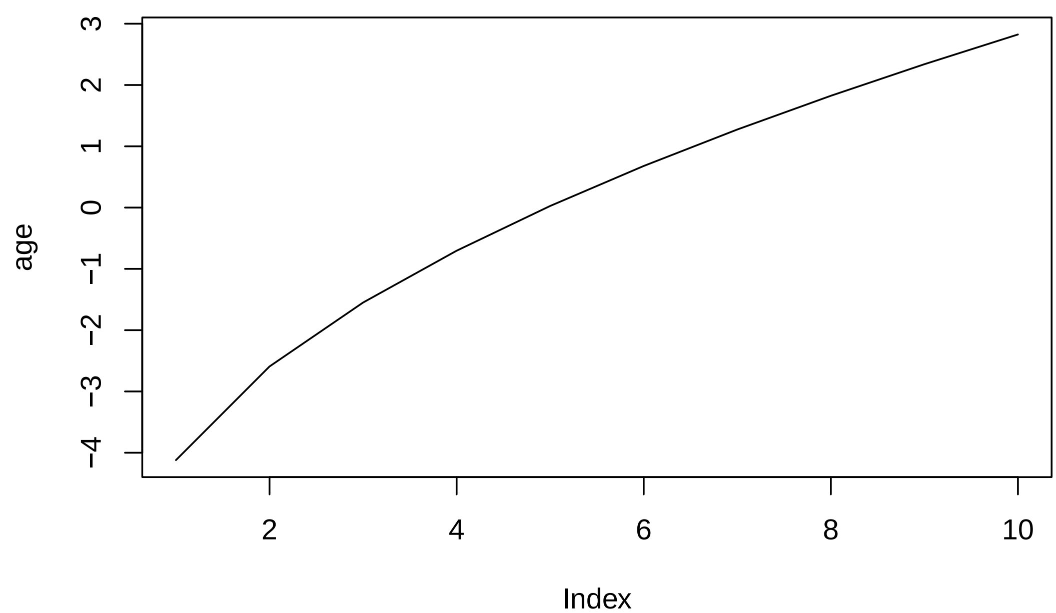
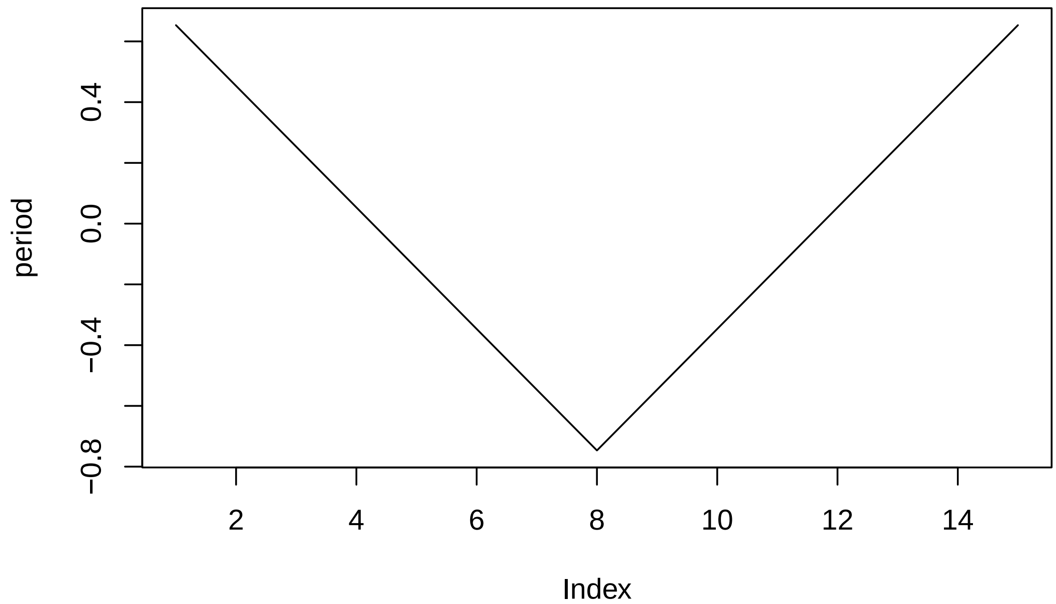
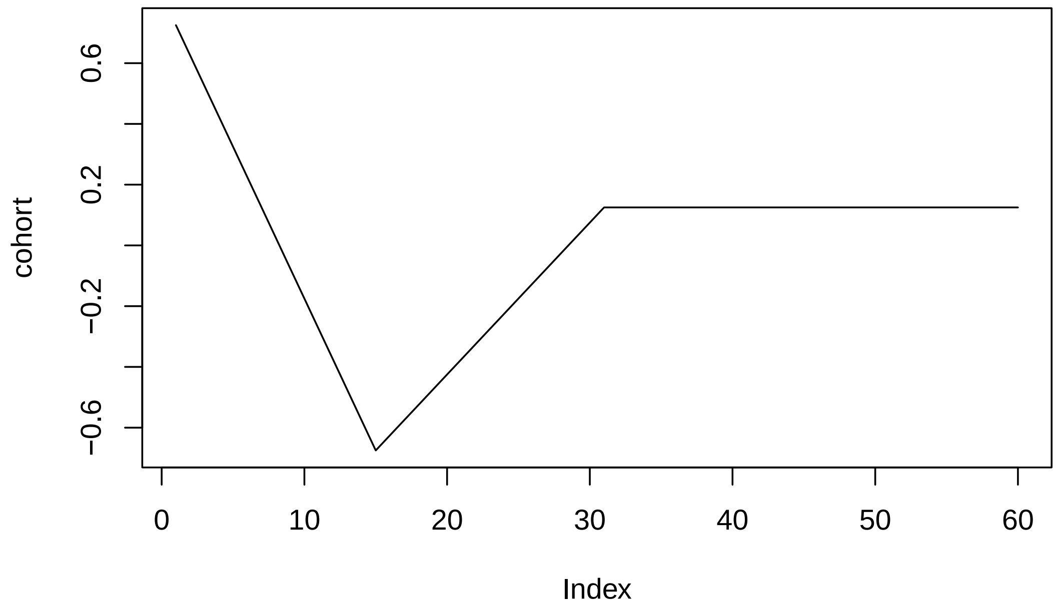
```
age=2*sqrt(seq(1,20,length=10))
age<- age-mean(age)
plot(age, type="l")
```




```
period=15:1
period[8:15]<-8:15
period<-period/5
period<-period-mean(period)
plot(period, type="l")
```




```
periods_per_agegroup=5
number_of_cohorts <- periods_per_agegroup*(10-1)+15
cohort<-rep(0,60)
cohort[1:15]<-(14:0)
cohort[16:30]<- (1:15)/2
cohort[31:60]<- 8
cohort<-cohort/10
cohort<-cohort-mean(cohort)
plot(cohort, type="l")
```




```
simdata<-apcSimulate(-10, age, period, cohort, periods_per_agegroup, 1e6)
print(simdata$cases)
```

```
##       [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10]
##  [1,]    3    4   15   49   76  114  158  388 1170  3135
##  [2,]    2    2   25   49   75  109  136  328  877  2263
##  [3,]    1    5   15   35   59   99  135  238  613  1646
##  [4,]    0    5   18   27   49   88   98  164  470  1228
##  [5,]    1    8    6   21   35   73   96  136  360   958
##  [6,]    1    5    4   19   37   50   73  108  237   633
##  [7,]    0    3    6   16   20   44   65   94  175   508
##  [8,]    1    0    3   11   30   44   55   70  149   361
##  [9,]    0    6    5   17   25   52   64   94  153   460
## [10,]    1    1    5   18   36   86  116  107  174   422
## [11,]    1    3    5   26   52   75  121  172  197   515
## [12,]    0    4    9   24   65  107  146  207  306   582
## [13,]    2    5   17   35   80  137  201  255  409   600
## [14,]    0    3   10   39   84  177  239  341  479   718
## [15,]    0   10   16   51  104  217  322  478  585   694
```

```
simmod <- bamp(cases = simdata$cases, population = simdata$population, age = "rw1",
period = "rw1", cohort = "rw1", periods_per_agegroup =periods_per_agegroup)
```

```
print(simmod)
```

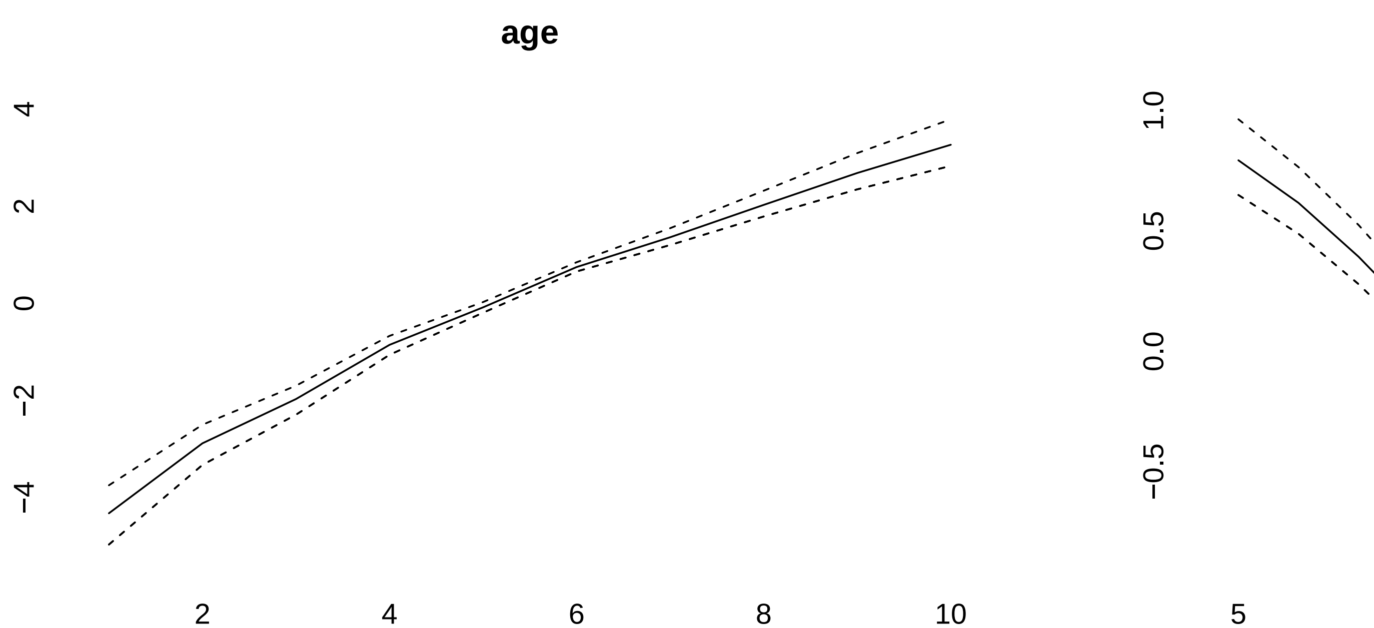
```
##
##  Model:
```

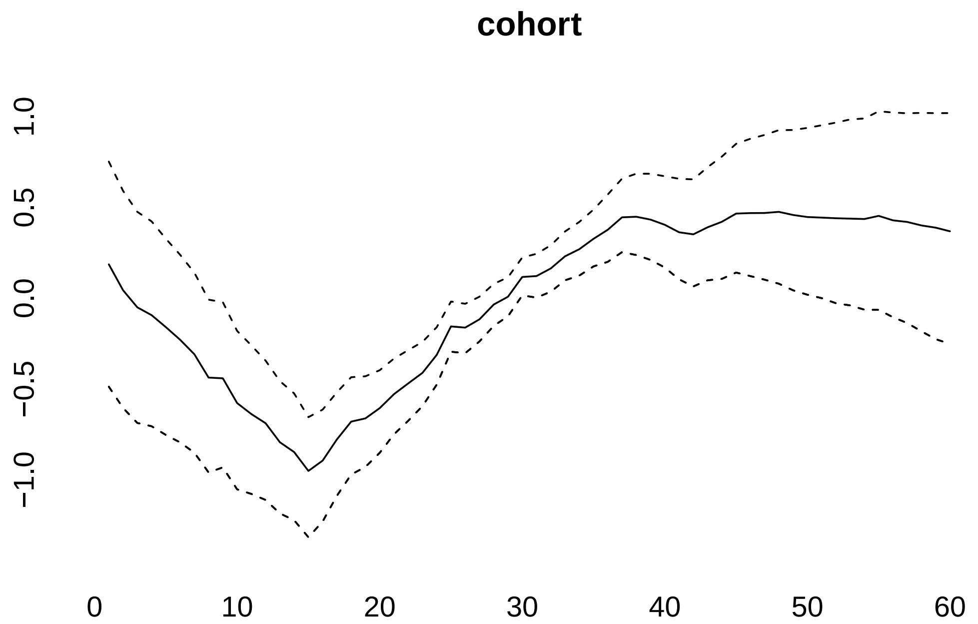
```
## age (rw1)  - period (rw1)  - cohort (rw1) model
## Deviance:     153.73
## pD:            49.83
## DIC:          203.55
##
##
##  Hyper parameters:                    5%          50%          95%
## age                               0.548        1.247        2.522
## period                           13.243       25.728       44.901
## cohort                           72.907      113.581      173.855
##
##
## Markov Chains convergence checked succesfully using Gelman's R (potential scale reduction factor).
```

```
checkConvergence(simmod)
```

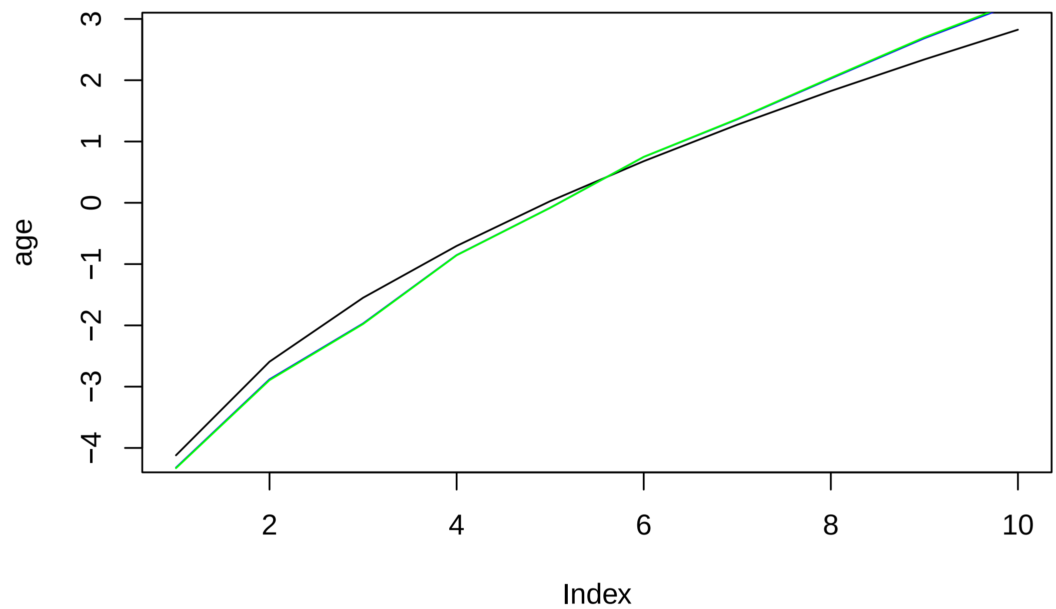
```
## [1] TRUE
```

```
plot(simmod)
```

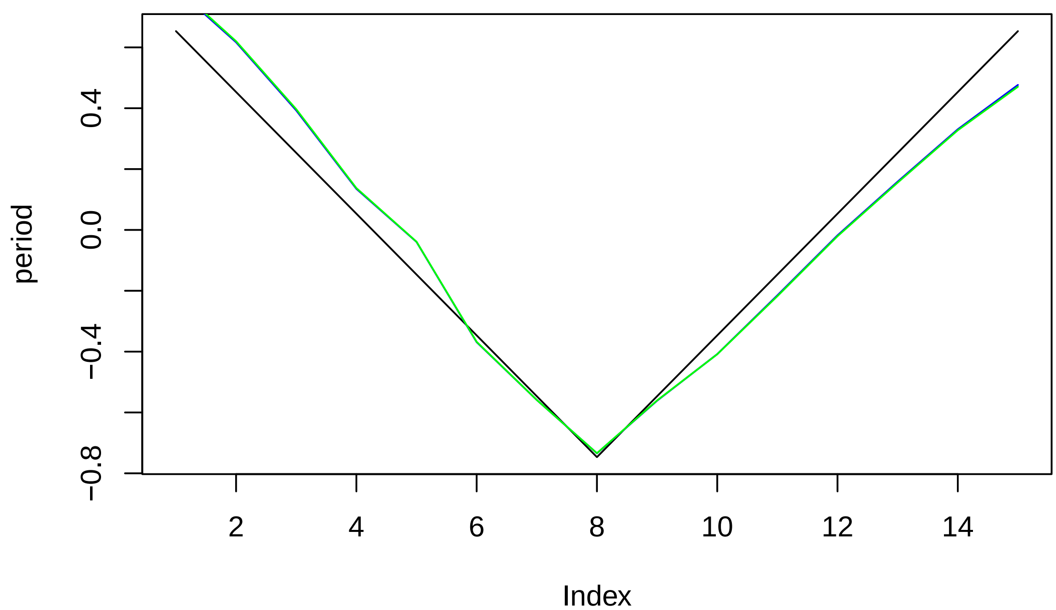




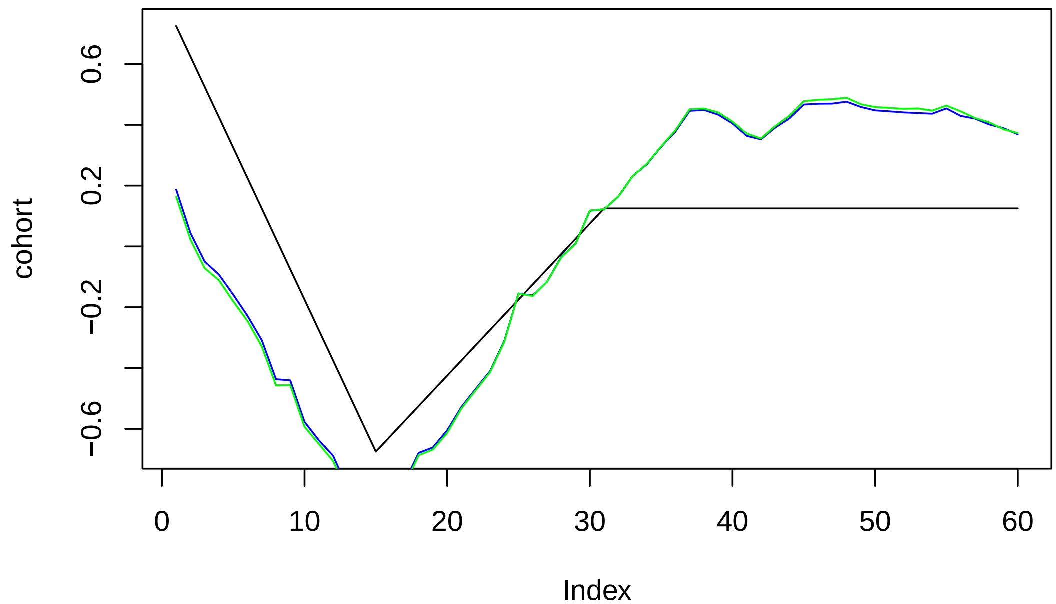

```
effects<-effects(simmod)
effects2<-effects(simmod, mean=TRUE)
#par(mfrow=c(3,1))
plot(age, type="l")
lines(effects$age, col="blue")
lines(effects2$age, col="green")
```




```r
plot(period, type="l")
lines(effects$period, col="blue")
lines(effects2$period, col="green")
```

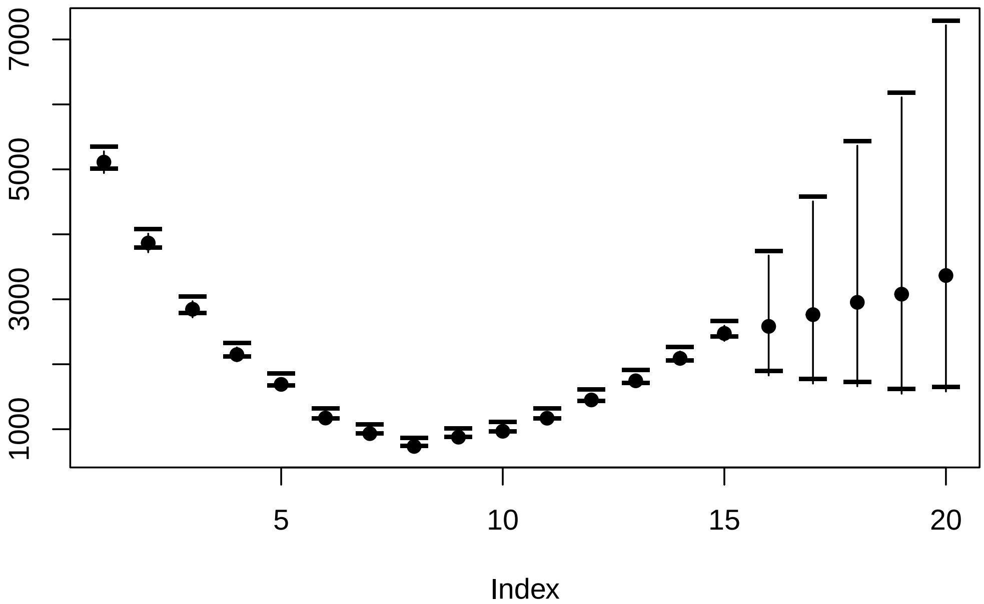


```r
plot(cohort, type="l")
lines(effects$cohort, col="blue")
lines(effects2$cohort, col="green")
```

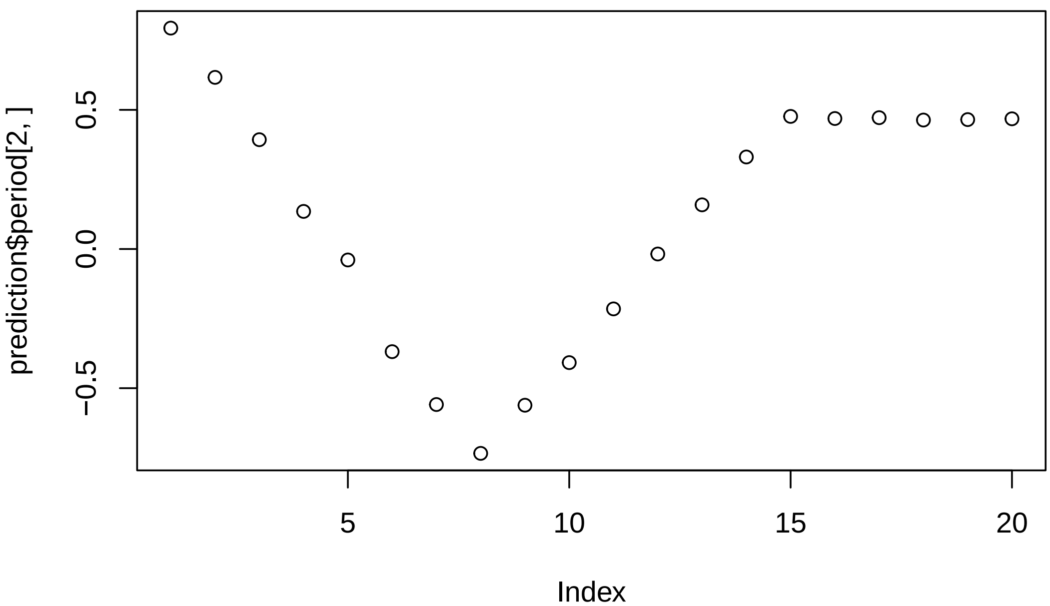



```r
prediction<-predict_apc(simmod, periods=5, population=array(1e6,c(20,10)))
```

```r
plot(prediction$cases_period[2,], ylim=range(prediction$cases_period),ylab="",pch=19)
points(prediction$cases_period[1,],pch="-",cex=2)
points(prediction$cases_period[3,],pch="-",cex=2)
for (i in 1:20)lines(rep(i,3),prediction$cases_period[,i])
```




```r
plot(prediction$period[2,])
```



```r
cov_p<-rnorm(15,period,.1)
```

```r
simmod2 <- bamp(cases = simdata$cases, population = simdata$population, age = "rw1",
period = "rw1", cohort = "rw1", periods_per_agegroup =periods_per_agegroup,
period_covariate = cov_p)
```

```
## Warning: MCMC chains did not converge!
```

```r
print(simmod2)
```

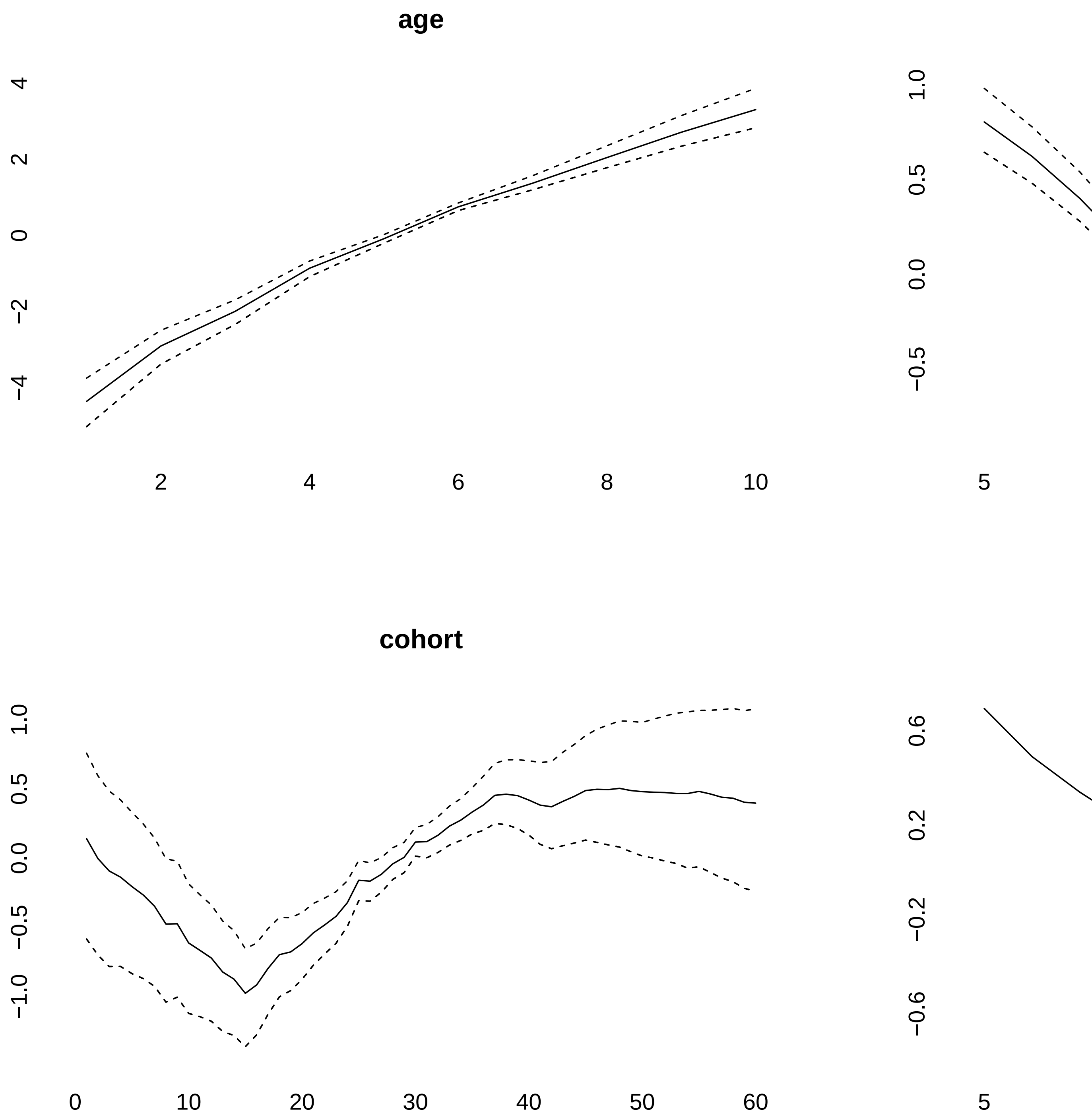
```
##
## WARNING! Markov Chains have apparently not converged! DO NOT TRUST THIS MODEL!
##
##  Model:
## age (rw1)  - period (rw1)  - cohort (rw1) model
## Deviance:     153.75
## pD:            49.85
## DIC:          203.60
##
##
##  Hyper parameters:                   5%          50%          95%
## age                               0.534        1.246        2.505
## period                           12.700       25.293       43.959
## cohort                           71.599      113.538      172.393
```

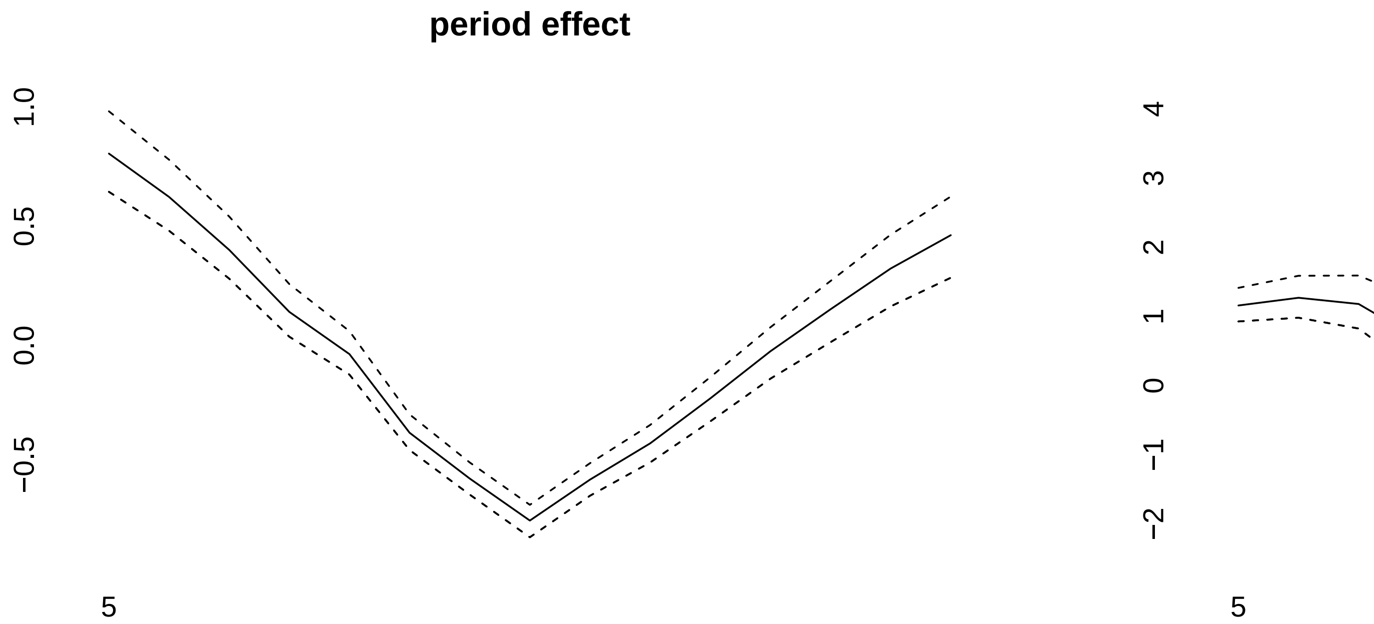
```r
checkConvergence(simmod2)
```

```
## Warning: MCMC chains did not converge!
```

```
## [1] FALSE
```

```r
plot(simmod2)
```


age
4
2
0
−2
−4
2
4
6
8
10
1.0
0.5
0.0
−0.5
5
cohort
1.0
0.5
0.0
−0.5
−1.0
0
10
20
30
40
50
60
0.6
0.2
−0.2
−0.6
5


period effect
1.0
0.5
0.0
−0.5
5
4
3
2
1
0
−1
−2
5